

PET and Gestational Hypertension 08: Delivery, Induction, and the Fetus

Companion reading handout for simultaneous listening.

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Delivery is the only definitive treatment for pre-eclampsia, but that sentence is too simple to manage a real patient.

The better question is this: when does birth become safer than continuing the pregnancy?

That question has three parts. What is the maternal risk of waiting? What is the fetal and placental risk of waiting? And what is the neonatal risk of birth now?

At term, the answer is usually straightforward.

For gestational hypertension or pre-eclampsia without severe features, delivery is recommended at thirty seven weeks or later. This is not arbitrary. The HYPITAT trial randomized women with gestational hypertension or mild pre-eclampsia after thirty six weeks to induction or expectant monitoring. Poor maternal outcome occurred in thirty one percent with induction versus forty four percent with expectant management. No maternal or neonatal deaths or eclampsia were recorded. The trial supports induction at term because maternal risk falls and neonatal cost is low.

So at thirty seven weeks with non-severe disease, the question is not, can we watch a little longer? The question is, what does watching still buy?

Before thirty seven weeks without severe features, the balance is different. Fetal maturity still matters. If BP is non-severe, labs are stable, symptoms are absent, and fetal assessment is reassuring, expectant management with close surveillance can be appropriate.

NICE expresses this conservatism clearly for gestational hypertension: if BP is below one hundred sixty over one hundred ten, do not offer planned early birth before thirty seven weeks unless other medical indications develop. That statement is useful because it prevents overreaction to a label. Hypertension alone, if non-severe and stable before thirty seven weeks, is not automatically a delivery indication. The delivery indication comes from gestational age, severe BP, maternal organ involvement, fetal compromise, or a worsening trajectory.

Severe Features

Now severe features.

At thirty four weeks or beyond, ACOG recommends delivery after maternal stabilization for pre-eclampsia or gestational hypertension with severe features. Do not delay indicated delivery just to complete late-preterm steroids.

Before thirty four weeks, expectant management is possible only for selected stable patients in centers that can care for very preterm infants and sick mothers. The word "stable" is doing heavy work. The mother must be stable. The fetus must be stable enough. The team must be able to monitor intensively and deliver quickly.

There are indications for delivery at any gestational age. Uncontrolled severe BP despite therapy. Eclampsia. Stroke. Persistent severe neurologic symptoms. Pulmonary edema. Worsening renal function. Progressive thrombocytopenia. HELLP syndrome. DIC. Abruptio. Nonreassuring fetal status. Fetal demise. Severe fetal compromise such as reversed end-diastolic flow depending on the full context and local protocol.

The 34 to 36+6 Week Window

Now the difficult window: thirty four to thirty six plus six.

This is where guidelines sound different because the tradeoff is real.

PHOENIX studied planned delivery versus expectant management for late-preterm pre-eclampsia from thirty four to thirty six plus six. Planned delivery reduced maternal morbidity and severe hypertension. It increased neonatal unit admission, mainly because babies were born earlier, without a clear increase in severe neonatal morbidity. The two-year follow-up did not show a clear clinically important harm signal, but it also could not prove noninferiority for every neurodevelopmental measure.

So the teaching phrase is this: in late-preterm pre-eclampsia, delivery buys maternal safety and spends neonatal maturity.

If the mother has severe features that mandate delivery, the decision is made. If she is stable and the fetus is stable, the decision should include gestational age, steroid exposure, neonatal resources, maternal trajectory, fetal status, and patient values.

NICE handles this late-preterm window by emphasizing senior decision-making. From thirty four to thirty six plus six, planned early birth is considered when maternal or fetal indications exist, when antenatal corticosteroids have been considered, and when the neonatal team has been involved. That is the right tone. The resident should not treat this as an automatic calendar rule. It is a structured conversation with obstetrics, maternal-fetal medicine when available, anesthesia, neonatology, and the patient.

Mode of Delivery

Now mode of delivery.

Pre-eclampsia is not an automatic cesarean indication.

ACOG states that for gestational hypertension or pre-eclampsia without severe features, vaginal delivery is preferred unless there is an obstetric indication for cesarean. Even in severe disease, the route depends on maternal stability, fetal status, gestational age, presentation, cervical status, and urgency.

At term or near term, induction is often appropriate. If the cervix is favorable, amniotomy and oxytocin may be efficient. If the cervix is unfavorable, cervical ripening is a logistics problem, not a reason to ignore the delivery indication. Mechanical ripening, prostaglandins, or combined methods depend on local protocol, uterine scar status, and fetal condition.

If there is prior cesarean, uterine surgery, severe fetal compromise, abruption, unstable eclampsia, or maternal decompensation, the delivery route may change. The principle is not vaginal at all costs. The principle is route of delivery chosen for the safest timely birth.

Induction planning should be honest about time. A nulliparous patient at thirty seven weeks with an unfavorable cervix may need cervical ripening, patience, and careful monitoring. A patient with worsening platelets, escalating antihypertensive needs, pulmonary edema, or a deteriorating fetal tracing may not have that time. The cervix matters because it predicts how long induction may take, not because it cancels the medical need for delivery.

During induction, continue to treat severe BP promptly. Continue magnesium when indicated. Avoid unnecessary fluid loading. Involve anesthesia early, especially if platelets are falling. Watch urine output without reflexively bolusing fluid for every low number. Delivery planning and maternal stabilization are simultaneous, not sequential chores.

The Fetus

Now the fetus.

The fetus is not separate from the disease. In early pre-eclampsia especially, the fetus may be showing the placental side of the same pathophysiology. Fetal growth restriction, oligohydramnios, abnormal umbilical artery Doppler, abruption, and stillbirth risk are all part of the placental disease spectrum.

This is why fetal surveillance is not just NST checkboxes.

At diagnosis, assess fetal growth. Assess amniotic fluid. Use umbilical artery Doppler when growth restriction or placental insufficiency is suspected. At early gestations

with severe fetal growth restriction, ductus venosus Doppler may help where expertise exists. ISSHP recommends against relying on the biophysical profile in hypertensive and growth-restricted pregnancies because it can be falsely reassuring and become abnormal late.

The resident must learn this: a normal BP trend does not guarantee placental stability. Maternal and fetal trajectories can diverge. A mother may look stable while the fetus declares placental failure. Or fetal testing may look acceptable while the maternal brain, lungs, kidneys, liver, or platelets become the reason to deliver.

This is also why fetal testing cannot be used to overrule maternal danger. A reassuring tracing does not make pulmonary edema safe. Normal growth does not make eclampsia less urgent. Good fetal status may help choose route and timing over the next hours, but it does not erase severe maternal disease.

Anesthesia

Now anesthesia.

Early anesthesia involvement is part of good care. Neuraxial anesthesia is often valuable because it avoids airway edema, aspiration risk, and the hypertensive response to intubation. ACOG states neuraxial techniques are generally acceptable when platelets are at least seventy times ten to the ninth per liter, stable, and there is no coagulopathy or concerning anticoagulant issue.

General anesthesia in severe pre-eclampsia is not just routine general anesthesia. Airway edema can make intubation difficult. Intubation and extubation can spike BP. Severe hypertension during airway manipulation can trigger stroke.

Finally, eclampsia and fetal tracing.

After a seizure, fetal bradycardia and uterine hyperactivity may occur. Gabbe emphasizes that these often resolve after maternal stabilization. Do not rush to cesarean solely because the tracing is abnormal during the seizure. Stabilize the mother, treat BP, give magnesium, oxygenate, reposition, then reassess.

Closing Synthesis

Let us close with the delivery equation.

Maternal risk of continuing, plus fetal and placental risk of continuing, versus neonatal risk of birth now.

At thirty seven weeks without severe features, birth usually wins. At thirty four weeks or beyond with severe features, birth usually wins after stabilization. Before thirty four weeks, time can be valuable, but only if maternal and fetal trajectories remain stable in the right setting.

Resident Reflex

The resident reflex is this: delivery timing is not a date memorized from a table. It is risk arithmetic, anchored by HYPITAT at term, PHOENIX in the late-preterm window, and maternal stabilization whenever severe features appear.